Technical Writing Laboratory Portfolio

Overview of Tools and Standards

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Introduction

Technical writing involves presenting specialised information in a clear, structured, and effective manner.

This presentation introduces five important tools used in technical documentation:

1. Git and GitHub
2. LaTeX
3. Beamer
4. Doxygen
5. Google Docs

Git and GitHub

Git

- A distributed version control system.
- Tracks changes in project files.
- Supports branching, merging, and commits.

GitHub

- An online platform for hosting Git repositories.
- Supports collaboration and code review.
- Provides Pull Requests for proposing changes.

Basic Git Workflow

A typical Git workflow consists of the following commands:

Essential Commands

- `git add` – Stage changes.
- `git commit` – Save a snapshot.
- `git push` – Upload commits.
- `git pull` – Retrieve and integrate changes.

Version control helps maintain project history and manage collaborative development.

LaTeX: Scientific Typesetting

LaTeX is a document preparation system widely used for academic and technical publications.

Key Features

- Structured document formatting.
- Mathematical equations and symbols.
- Automatic numbering of sections and figures.
- Tables, references, and bibliographies.

Example Mathematical Expression

$$E = mc^2$$

LaTeX Beamer

Beamer is a LaTeX document class used to prepare professional presentation slides.

Important Features

- Frames for individual slides.
- Sections and subsections for organisation.
- Themes and colour customisation.
- Mathematical expressions and lists.

Basic Beamer Structure

```
\begin{frame}
Slide content
\end{frame}
```

Doxygen

Doxygen is a documentation generation tool commonly used for C and C++ projects.

Features

- Extracts documentation from source comments.
- Generates HTML and other documentation formats.
- Organises functions, classes, and files.

Common Documentation Tags

`@file` – File description
`@brief` – Short description
`@param` – Function parameter
`@return` – Return value

Google Docs

Google Docs is an online word-processing tool that supports collaborative document creation.

Important Features

- Real-time collaborative editing.
- Heading styles and document formatting.
- Comments and suggestion mode.
- Automatic cloud saving.
- Exporting documents as PDF files.

Conclusion

The tools discussed provide an integrated approach to technical writing and documentation.

- Git and GitHub manage project versions.
- LaTeX produces structured technical reports.
- Beamer creates professional presentations.
- Doxygen generates software documentation.
- Google Docs enables collaborative writing.

Thank You